From the brainstorming to the prototype

Requirements engineering in a green field investment

Gábor Árpád Németh

Requirements engineering in a green field investment 8/1

From the brainstorming to the prototype

1. Initial thoughts - Brainstorming
   - What we would like to achieve?
   - Define scope
   - Define possible customers
   - Define initial budget (and maybe some forecast for later stages)

2. Refine requirements
   - Risks:

Too general

→ too complex architecture → too complicated to implement a simple functionality
→ most of the possible functionalities will never be used

vs.

Too specific

→ bad architecture → dirty hacks over hardwired structure → hard to maintain
→ no future-proof (lack of important features)

Requirements engineering in a green field investment 8/2
From the brainstorming to the prototype

2. Refine requirements

- We should get known possible customers, their goals, problems and current processes…etc.
  - Otherwise: Ivory tower: we make something that nobody really wants…

An example: Ikarus PALT*:

A big engineering achievement that was not applicable to the existing infrastructure

Source of figures: Ikarus archives *PALT: Passengers And Luggage Together

Requirements engineering in a green field investment 8/3
From the brainstorming to the prototype

- An example of collecting requirements – in a structured document:

Requirement Specification of Tool Zebra

Contents

1 The Zebra system: our vision 4
1.1 The philosophy of the tool Zebra 4
1.2 Terminology 4
1.2.1 Behavior 4
1.2.2 Topology 4
1.2.3 User 6
1.2.4 Zebra 6
1.2.5 Zebroid 7
2 Requirements 7
2.1 Topology and behavior views 8
2.2 Setting of protocols and interfaces 13
2.3 Setting of user rights 17
2.4 Setting of behavior dependencies 19
2.5 Report generation 21
2.6 Miscellaneous 25
3 References 25
4 Abbreviations 25

1.2 Terminology

1.2.1 Behavior

The behavior is a dynamic description. It describes the interaction between the different participants of the system.

1.2.2 Topology

The topology is a static description, which defines the participants of the system and the interconnections among them. The parts of the Topology are described in Figure 1.

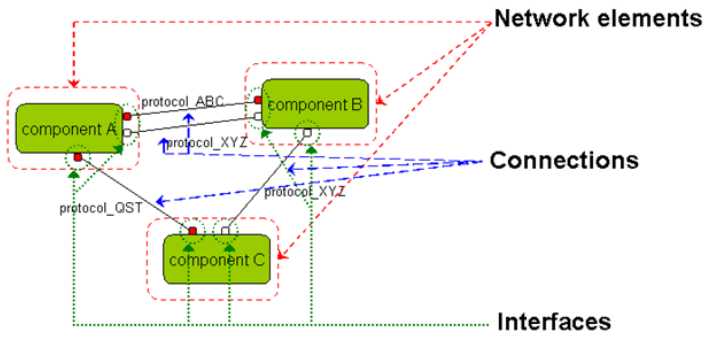


Figure 1: Parts of the topology

1.2.2.1 Network Elements

- The Network Elements are the nodes of the Topology. They are the participants of the interaction in the Topology.

R_G_20

Handle different abstraction levels.

One of the best properties of the engineer is the ability to select the right amount of details for a given task. It means that (s)he can consider a higher abstraction level if required, and not confused by any unimportant details. Thus, the tool should support the visualization and the editing in different abstraction levels. The tool should give the opportunity to the user to fold and unfold the given abstraction level on demand, to hide the details, which he is not interested in at a given time.

There are many requirements, which are related to this high-level requirement, see requirements R_G_21, R_U_05, R_U_06, R_T_03 and R_T_04 for further details.

R_G_21

Handle different abstraction levels of the topology. The tool should support to fold and unfold the different parts of the topology on demand.

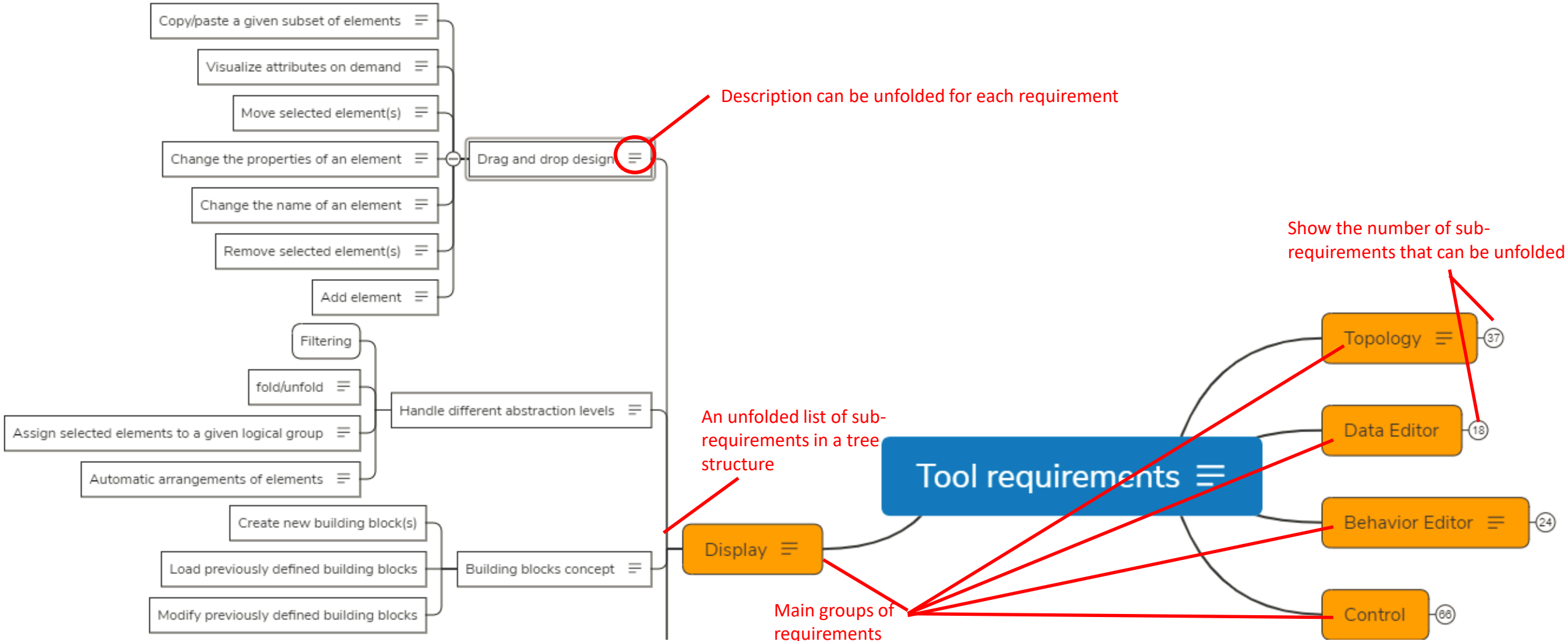
The user wants to display only that part of the network, which he is interested in. He wants to create higher and lower level view on demand.

An implementation idea would be if

(1) the user double clicks to an item, which he wants to unfold – for instance a given abstract

Requirements engineering in a green field investment 8/4
From the brainstorming to the prototype

- An example of collecting requirements – tree structure (xmind)



Requirements engineering in a green field investment 8/5
From the brainstorming to the prototype

3. Create a prototype
   - **Only for proof of concept!**

   - Should answer the following questions (2/1):
     - **What** would we like to achieve?
       - List of functionalities
     - **How** would we like to achieve the goal? – non-functional aspects
       - Usability ↔ user interface, assumptions about users, working process
       - Performance related aspects:
         - Responsibility
         - Designed workload
         - Scalability ↔ software architecture & required hardware
         - How to handle overload…etc.
       - Security aspects ↔ architecture

Requirements engineering in a green field investment 8/6

From the brainstorming to the prototype

3. Create a prototype
   - Should answer the following questions (2/2):
     - How we should provide expected **quality**?
       - Manual testing for explanatory testing
       - A few proof-of-concept tests
         - Unit, integration, system levels
         - Functional and non-functional (performance, (G)UI, security…etc.)

Requirements engineering in a green field investment 8/7

From the brainstorming to the prototype

- Prototype:
  - **Role: Proof-of-concept**
  - Not an implementation code base for the final product!

- From most of the prototypes no real product has been developed due to the following reasons:
  - Wrong assumptions, when defining requirements and scope
  - Wrong initial thoughts about possible customers and/or their needs
  - Wrong assumptions about budget
  - The protype showed that the development cost and/or time would be too high
  - Organizational changes in the company resulted in cost cut / project closure
  - Similar product has been developed meanwhile in parallel

Requirements engineering in a green field investment 8/8
From the brainstorming to the prototype

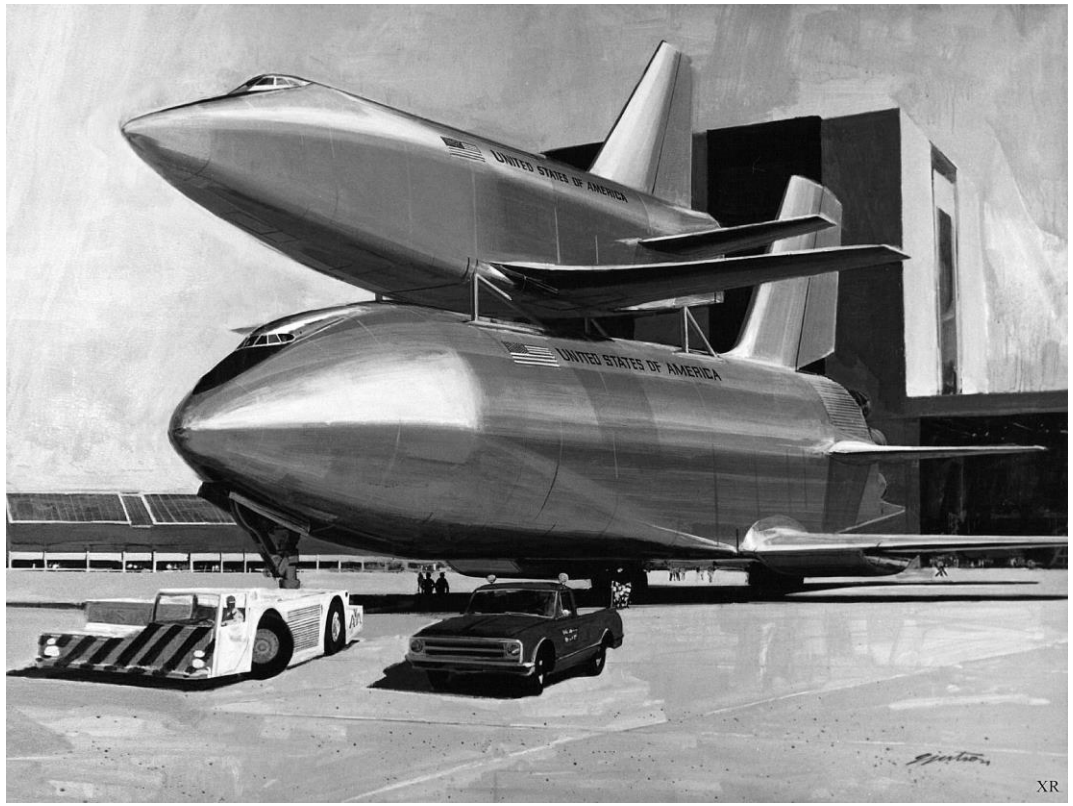
If we succeed than comes…

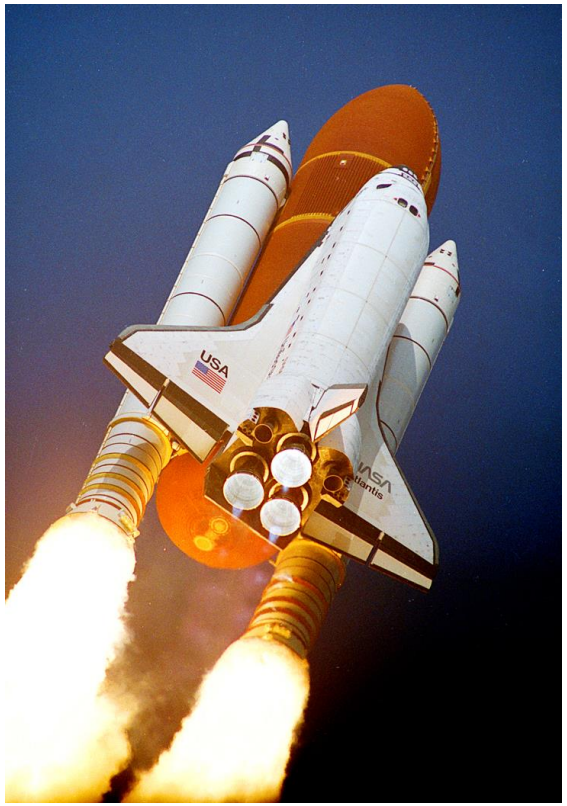
4. Productification
   - But many reviews before this step:
     - Technical reviews (architecture)
     - Management reviews at different levels (financial, customer…etc. aspects)

Project vs. product

	Project	Product
Generic	Unique, customer specific	Generic
Time	Has beginning and end date	Permanent (until phase out)
Planning	One-step/Predictive planning	Iterative/adaptive planning
Input	Project requirements	Evolving customer needs

A case study about changing requirements 9/1





The true story of the genesis of the Space Shuttle

References:

- David Baker: NASA Space Shuttle. 1981 onwards (all models). Owner's Workshop Manual. Haynes. 2011.
- Wikipedia: Space Shuttle program, Space Shuttle design process, Criticism of the Space Shuttle program

A case study about changing requirements 9/2

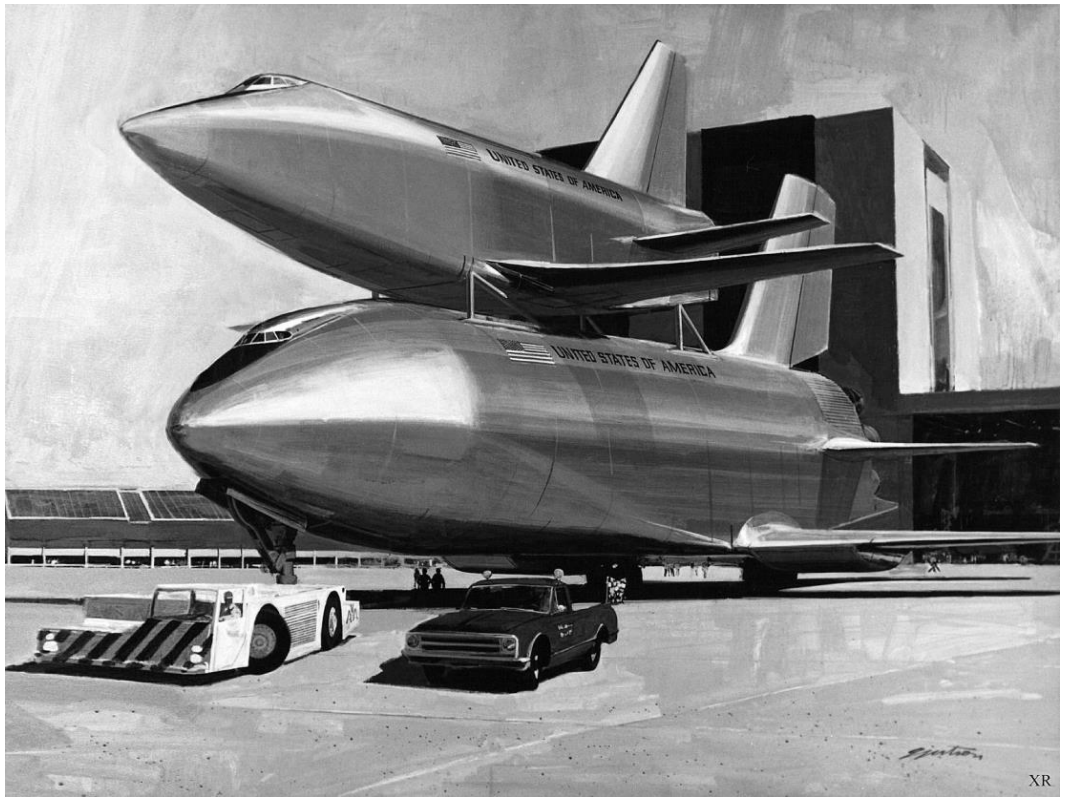
Background:

- After the Apollo (Moon landing program), significant cut on NASA's budget
- NASA plan to develop a fully reusable system – a „Shuttle" – to make space travelling significantly cheaper

Initial design (1970 Phase B studies):

- Fully reusable system:
  - 1.700 ton fly back manned booster with 12 rocket engines (with liquid fuel) and wings
  - 380 ton orbiter with 2 rocket engines
  - Orbiter's Payload capacity: 11 tons to LEO*

    * LEO: Low-Earth Orbit
- This solution was too costly to develop…



A case study about changing requirements 9/3

Investigate different concepts:

1. Fully (booster + orbiter) reusable systems
2. Expandable tanks
3. Expandable boosters

…etc

Catch-22:

- Lowest cost-per-flight solutions requires highest development cost
  - NASA have insufficient budget
- If decrease development cost, it results in a higher cost-per-flight
  - Controversial to the initial goal

A case study about changing requirements 9/4

Optimizations:

1. Expandable tank has been selected to decrease development cost (smaller orbiter would be enough)

   → compromise: not fully reusable system

2. SRBs (solid rocket boosters) proposed instead of liquid propellant ones to decrease cost-per-flight
   - Advantages:
     - **Simple and cheap**
     - Much easier to handle, no fueling needs before launch
   - Disadvantage:
     - Less efficient than liquid propellant rockets
     - **Once ignited, can not be stopped** – 1st compromise on safety

     (NASA had a rule before to not use them for manned space flights)

3. Insufficient thrust can be gained from solid rockets to lift-off the entire system

   → instead of the usual serial concept, parallel concept is selected for solid rockets

4. Recoverable boosters proposed to decrease cost-per-flight

A case study about changing requirements 9/5

Wait, initial design has been altered!

Do we need the cheaper solid rocket boosters if they are reusable?

A case study about changing requirements 9/6

Compromise of partners / payload capacity:

- The development cost was still too high → NASA found a partner (USAF*) to share the costs
- What payload is required?**
  - NASA: 6,8 tons to LEO*** (for satellites)
  - USAF*: 18 tons to polar orbit ≈ 30 tons LEO*** (for military satellites)
  - NASA later: 20 tons to LEO*** (to build Freedom space station from modules)

* USAF: US Air Force

** **The most important question when designing the Space Shuttle…**

*** LEO: Low-Earth Orbit

A case study about changing requirements 9/7

Canceled Safety functions:

- Liftoff – designed due SRBs:
  - Blow out port for boosters to separate in case of failure during ascending

    → cancelled due weight
  - Abort solid rocket motors

    → cancelled due simplicity

- Landing:
  - Turbofan engines that pop out from a compartment of rear payload bay at landing

    → cancelled due weight and volume

A case study about changing requirements 9/8

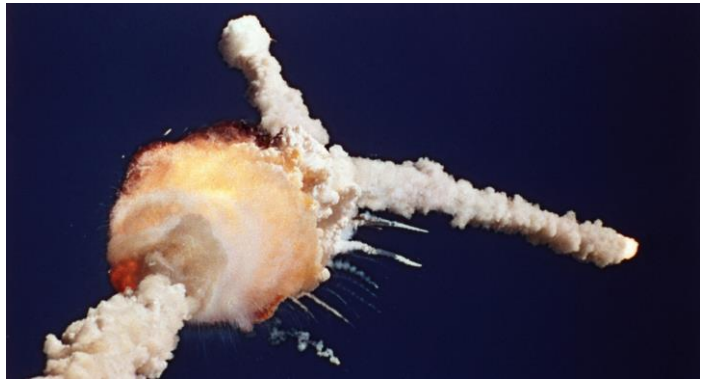
The usage of Space Shuttle - facts:

1. Due SRBs and parallel design, two Space Shuttle disasters:
   - Challenger in 1986

   → decrease flight intensity

   → US Air Force back out from project

   → increase cost-per-flight



   - Columbia in 2003

   → increase cost-per-flight

   → decision about the retirement of the Space Shuttle fleet

A case study about changing requirements 9/9

2. Except HST* in 1990, only after 1998 (17 years after first flight!) NASA uses the possible payload capability of the Space Shuttle, when building the ISS**

   * HST: Hubble Space Telescope

   ** ISS: International Space Station

3. The Vandenberg Space Shuttle launch pad build for US Air Force has never been used

4. The initial plan to send Space Shuttle into space bi-weekly has been never achieved

   → the cost-per-flight has been even higher compared to simple, expandable rockets!

5. After Space Shuttle era:
   - More simple designs
   - Concentrate on liquid propellant rocket engines
   - Concentrate on less payload capacity

Thank you for your attention!